Korea Technology and Information Promotion Agency for SMEs

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# Technology Proposal for 2026 TIPA* International Joint R&D Program

**(TIPA: Korea Technology and Information Promotion Agency for SMEs)**

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

## 1. Technology Proposer Information

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## 2. Technology Overview

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| **Title of Proposed Technology** | ***AI-Powered, Instruction-Driven Warehouse Robot Technology for Adaptive Material Handling*** | | | |
| **Korea's 12 National Strategic Technology Fields** | Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( ) | | | |
| **Industrial Technology Classification** 1) | **Technology Field 1** (Large Category) | | **Technology Field 2** (Large Category) | |
| | Mechanical · Materials | | Mechanical · Materials | |
| | Medium Category | Small Category | Medium Category | Small Category |
| | Robotics / Automation Machinery | Robot Control and Intelligence Technologies | Industrial / General Machinery | Transportation / Material Handling Machinery |
| **Concept of Proposed Technology** | The proposed technology aims to develop an **AI-powered, instruction-driven autonomous warehouse material handling robot system** that enables robots to flexibly and robustly perform material handling tasks (e.g., lifting, placing, and stacking) directly instructed by humans, rather than relying on rigid, pre-programed rules. This technology will allow the robot to **adapt to changing warehouse layouts, materials, and operational conditions** without extensive reconfiguration.<br><br>Current autonomous warehouse robots heavily rely on *prior knowledge* of environments and materials (e.g., shapes and dimensions) and on rule-based planning and control pipelines, often | | | |

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| | requiring labor-intensive engineering and lengthy deployment workflows to adapt to new environments or material variations. Unlike rule-based systems built on separate perception, planning, and control modules, the proposed system leverages advanced vision-language-action (VLA) models to **jointly perceive 3D environments, interpret human language, and generate instruction-driven actions**. Using onboard sensors and natural-language commands (e.g., "move the wooden pallet on the left to the second shelf" or "place the container near the loading area"), the robot directly interprets human instructions and translates it into goal-directed navigation and manipulation behaviors. This approach largely reduces reliance on rigid, rule-based pipelines and enables highly adaptive and generalizable material handling operations across diverse warehouse environments.<br><br>To efficiently develop, validate, and deploy the VLA-powered robot system with **privacy consideration**, the proposed technology will be enabled by a Real-to-Sim-to-Real (R2S2R) development framework. This framework constructs **physics-aware, photorealistic digital twins of the target warehouse** from real-world sensor data, enabling large-scale training and testing of VLA models in simulation prior to deployment. Once validated, the trained policies can be **zero-shot deployed** to real robotic platforms in the target warehouse, thereby reducing deployment risk, preserving data privacy, and minimizing on-site engineering effort.<br><br>The proposed system is designed to be scalable and robot-agnostic, supporting a wide range of warehouse material handling tasks such as navigation, pallet pickup, placement, and transfer. It can be adapted to diverse warehouse layouts, material types, and operational workflows, making it suitable for logistics centers, smart factories, manufacturing, and distribution facilities.<br><br>Key features:<br>● Vision-language-based, instruction-driven perception and reasoning for autonomous warehouse material handling;<br>● Physics-aware, photorealistic warehouse digi for zero-shot sim-to-real policy transfer;<br>● Privacy-preserving model development and deployment enabled by Real-to-Sim-to-Real framework;<br>● Adaptive operation across diverse warehouse layouts, material types, and operational conditions. |
| **Rationale for Joint R&D Project** | Autonomous material handling robots, including autonomous mobile robots (AMRs), automated guided vehicles (AGVs), and autonomous forklifts, play an essential role in modern logistics and manufacturing. Nevertheless, rapidly adapting these systems across diverse deployment environments remains a major technical barrier that requires advanced research and development efforts. For SMEs, this barrier leads to high labor costs, increased deployment risk, and extended operational downtime. |

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| | In the meantime, recent advances in vision-language foundation models present new opportunities for enabling more flexible, adaptive, instruction-driven material handling robots. However, developing such systems necessitates the integration of advanced AI models with real warehouse robotic platforms, realistic warehouse environments, and domain-specific expertise, which is challenging for a research institute to achieve independently. Therefore, joint R&D between a research institute and an industry partner is essential to project success and technology transfer.<br><br>Rationale for collaboration:<br>● The research institute contributes advanced vision-language-action models, simulation technologies, and autonomy algorithms;<br>● The industry partner provides real-world industrial requirements, robotic hardware platforms, operational expertise, and access to representative warehouse environments;<br>● Effective co-development, validation and refinement of AI software and robotic hardware is necessary to meet real-world industrial needs, reduce technical risk and accelerate technology transfer<br><br>Expected benefits from joint R&D:<br>● Accelerated technology transfer from AI research to deployable warehouse robots;<br>● Reduced technical and deployment risk through joint development and real-world validation;<br>● Improved industrial relevance by integrating advanced research with operational requirements;<br>● Creation of a scalable foundation for commercialization of AI-powered warehouse robots. |
| **Key Words** | 1. Vision-Language-Action-based Material Handling<br>2. Autonomous Warehouse Robots<br>3. Adaptive Warehouse Automation<br>4. Real-to-Sim-to-Real Framework<br>5. Physics-Aware Warehouse Digital Twins |

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## 3. Research and Development Objectives & Plan

| Technology Readiness Level (TRL) [2] | Start of the Project | Level ( 3 ) | End of the Project | Level ( 6 ) |
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| **R&D Objectives** | The objective of this R&D project is to develop an AI-powered, instruction-driven, and adaptive autonomous warehouse material handling robot system capable of flexibly and reliably executing human-instructed material handling tasks across diverse warehouse environments through integrated perception, language understanding, and action generation.<br><br>The specific R&D objectives are:<br><br>● To develop a Real-to-Sim-to-Real framework for constructing privacy-preserving, physics-aware, and photorealistic warehouse simulation environments from real-world sensor data;<br>● To develop, train, and validate vision-language-action models for warehouse material handling tasks such as pallet lifting, placing, and stacking;<br>● To achieve reliable zero-shot transfer of trained policies from simulation to real-world warehouse environments. | | | |
| **R&D Target Specifications** | The target performance specifications of this R&D project are as follows:<br><br>**Instruction Understanding and Task Execution**<br>● Robust 3D perception and language understanding with ≥ 80% target detection and localization accuracy for common warehouse objects (e.g., pallets, containers, racks) under varying lighting and layout conditions.<br>● Achieve ≥ 70% success rate in correctly executing high-level human instructions for material handling lifting and placing tasks in representative warehouse environments.<br><br>**Real-to-Sim-to-Real Framework**<br>● High-fidelity warehouse digital twin with ≤ 5 cm geometric reconstruction error, photorealistic rendering quality (PSNR ≥ 25 dB), and physically realistic interactions sufficient for VLA model training.<br>● Zero-shot policy transfer from simulation to real robotic platforms with ≤ 5% performance degradation relative to simulation results.<br><br>**System Adaptability and Reliability**<br>● Demonstrated adaptability across multiple warehouse layouts and material variants without retraining from scratch.<br>● Achieve ≥ 80% overall task success rate after policy fine-tuning in the target warehouse simulation environment. | | | |
| **R&D Plan** | This R&D project will be conducted over three years, following a progressive development and validation strategy from simulation to laboratory environments and finally to real-world warehouse settings. | | | |

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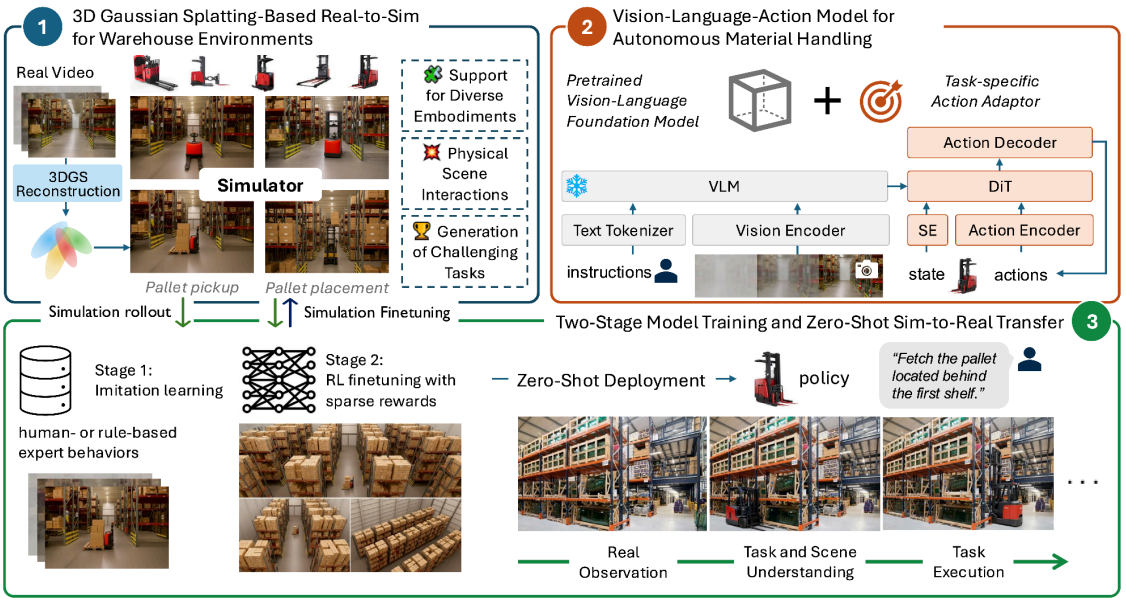


**Figure 1**. Overview of the development and validation strategy for the proposed autonomous warehouse material handling robotic technology.

- **Year 1: Real-to-Sim Digital Twin Construction and Simulation-based Development**
  **Objectives:** Establish a high-fidelity warehouse simulation environment and develop core AI algorithms in a risk-free, cost-effective manner
  **Key tasks:**
  - Develop a real-to-sim framework to construct physics-ware, photorealistic warehouse simulation environments that replicate physical warehouses from real-world sensor data, with 3D Gaussian Splatting (3DGS)-based visual rendering and realistic robot-object interactions
  - Design and implement a vision-language-action (VLA) model for autonomous warehouse robots to enable instruction-driven material handling
  - Collect large-scale training data (i.e., video observations, language instructions, and model-based action trajectories) in simulation for model training
  - Train and validate the developed VLA-based material handling policy in simulation and establish quantitative performance benchmarks

  **Milestones:**
  - High-fidelity warehouse digital twin that supports robot policy training and testing
  - Initial VLA model trained and evaluated on material handling tasks in simulation

- **Year 2: Laboratory-Scale System Integration and Refinement**
  **Objectives:** Integrate the developed AI algorithms with robotic hardware in a controlled lab environment to reduce deployment risk and enable rapid, iterative system refinement
  - Enhance the real-to-sim framework with 3D object generation and domain randomization to increase data diversity and improve model robustness
  - Integrate the sensors and AI algorithms onto a laboratory-scale warehouse robotic platform
  - Develop a human-robot instruction interface that allows operators to issue natural-language task commands
  - Conduct iterative training in augmented simulation and testing in laboratory environments to refine sensor configuration and model design

  **Milestones:**
  - Integrated AI-powered warehouse robotic system demonstrated in lab settings
  - A mature real-to-sim-to-real development workflow enabling rapid system iteration

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| | • **Year 3: Sim-to-Real Deployment and Technology Advancement**<br>**Objectives:** Demonstrate real-world applicability, scalability, and readiness for industrial adoption.<br>**Key tasks:**<br>○ Deploy the developed AI models on real warehouse robotic platforms operating in physical warehouse environments<br>○ Conduct large-scale hardware experiments to evaluate vision-language understanding, target object localization, and material handling task success rates<br>○ Demonstrate system adaptability by deploying the same robot across multiple warehouse environments with minimal reconfiguration<br>○ Refine system reliability, adaptability, and operational robustness based on experimental results<br>○ Assess commercialization potential and prepare the technology for industrial transfer, including documentation, system interfaces, and deployment guidelines<br>**Milestones:**<br>○ VLA-powered warehouse robot system demonstrated in real warehouse environments<br>○ Technology readiness advanced to TRL 6<br>○ Clear pathway identified for commercialization and SME adoption |

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## 4. R&R (Roles and Responsibilities)

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| **Expected R&R of Research Institute** | The research institute will serve as the primary technical lead for the joint R&D project, focusing on the development of advanced AI algorithms, simulation technologies, and system integration frameworks required for instruction-driven autonomous warehouse material handling.<br><br>The specific roles and responsibilities of the research institute include:<br>● Development of core AI technologies, including vision-language-action (VLA) models for integrated 3D perception, language understanding, and instruction-driven material handling;<br>● Development of the real-to-sim-to-real (R2S2R) framework, including 3D reconstruction, 3D Gaussian Splatting-based rendering, and physical interaction integration;<br>● Simulation-based development and validation, including algorithm benchmarking and laboratory-scale performance analysis of the proposed technology;<br>● Provide technical leadership, research resources, and support for technology transfer and deployment.<br><br>The Lunar Lab at Georgia Tech already has 2 Ph.D. students and 4 M.S. students actively working on the domain of warehouse material handling robots with highly promising preliminary results in simulation. |
| **Expected R&R of Korean SME Partner** | The Korean SME partner will play a critical role in ensuring the industrial relevance, practical feasibility, and successful deployment of the proposed technology by contributing real-world systems, operational expertise, and validation environments.<br><br>Specifically, the Korean SME partner will be responsible for:<br><br>● Providing robotic hardware platforms, sensors, and system integration support, such as autonomous mobile robots, forklifts, or other relevant warehouse robotic systems used in material handling operations;<br>● Defining real-world operational requirements and use cases, including warehouse layouts, material types, safety constraints, and task workflows, to guide system design and performance evaluation;<br>● Supporting data collection, system integration, testing, validation, and deployment activities in pilot-scale warehouse experiments, and performance evaluation in representative operational environments.<br>● Leading commercialization planning and industrial transfer, including assessment of deployment feasibility, cost considerations, and pathways for integrating the developed technology into existing warehouse automation solutions. |

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## 5. Expected Outcomes/Impact

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| **Expected Outcomes and Impact** | The proposed joint R&D project is expected to deliver an AI-powered, instruction-driven autonomous warehouse material handling robot system capable of flexibly and reliably executing human-instructed material handling tasks across diverse warehouse environments with minimal engineering effort. The project will advance **robotic physical intelligence, physics-aware simulation, and zero-shot sim-to-real transfer**, achieving measurable improvements in adaptability, robustness, and deployment efficiency compared to conventional rule-based systems.<br><br>From an industrial perspective, this project will provide the Korean SME partner with core AI-enabled material handling technologies and real-to-sim-to-real system integration know-how applicable to next-generation warehouse automation solutions, establishing a strong foundation for commercialization. The instruction-driven and adaptive system reduces deployment and reconfiguration costs, enabling faster adoption across diverse warehouse environments. The real-to-sim-to-real framework further supports scalable customization with minimal on-site engineering effort, making the technology well suited for **SME-led productization and service offerings**. More broadly, the project is expected to enhance the **global competitiveness of Korean SMEs in the warehouse robotics and logistics automation market** and contribute to productivity gains and digital transformation in logistics and manufacturing. |

## 6. References

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☐ **Terms:**

**1) Industrial Technology Classification**

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

**2) Technology Readiness Level (TRL)**

| **Description** | | |
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| ·A consistent and objective indicator of the technological maturity of key element technologies | | |
| | **Basic Research Stage (1-2)** | 1st Stage: Basic theories and experiments<br>2nd Stage: Proof of Concept, such as practical ideas or patents |
| | **Experimental Stage (3-4)** | 3rd Stage: Verification of basic performance at a laboratory scale<br>4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale |
| | **Prototype Stage (5-6)** | 5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems<br>6th Stage: Production and performance evaluation of prototypes at a pilot scale |
| | **Productization Stage (7-8)** | 7th Stage: Reliability testing and evaluation by potential customers<br>8th Stage: Prototype certification and standardization |
| | **Commercialization Stage (9)** | 9th Stage: Commercialization |